CYBERSECURITY PORTFOLIO PROJECT

SIEM Account Compromise Investigation

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Environment	Fictional healthcare organization
Focus	Authentication analysis, incident triage, containment, impact assessment
Tools/Concepts	SIEM, authentication logs, MFA, least privilege, incident response

1. Alert and Initial Analysis

The SIEM generated an alert after five failed logins for billing user **j.smith**, followed by a successful authentication from Germany. The employee normally signs in from Atlanta on a company Windows laptop. The successful session used an unknown Linux device and did not complete MFA.

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Authentication Alert - j.smith

MEDIUM ALERT

Time	User	Result / Event	Source / Details
08:41:03	j.smith	FAILED	10.20.5.18
08:41:17	j.smith	FAILED	10.20.5.18
08:41:29	j.smith	FAILED	10.20.5.18
08:41:46	j.smith	FAILED	10.20.5.18
08:42:02	j.smith	FAILED	10.20.5.18
08:44:16	j.smith	SUCCESS	185.220.101.42 | Germany | Unknown Linux

SIMULATED LAB EVIDENCE - No real patient or employer data

Figure 1. Simulated SIEM authentication alert showing repeated failures followed by an unusual successful login.

Initial assessment: The repeated failures, unusual geography, unknown device, and incomplete MFA created multiple indicators of possible unauthorized access.

2. Investigation and Verification

I reviewed the successful login's source IP, location, device, MFA status, authentication history, and activity after login. The account had no previous Germany logins. The employee confirmed the session was not theirs and that they were working in Atlanta.

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Post-Login Activity Timeline

LIKELY COMPROMISE

Time	User	Result / Event	Source / Details
08:44	j.smith	LOGIN SUCCESS	Germany | Unknown device | MFA incomplete
08:46	j.smith	PORTAL ACCESS	Employee portal
08:48	j.smith	SYSTEM ACCESS	Billing system
08:51	j.smith	RECORD SELECT	247 billing records
08:52	j.smith	DOWNLOAD	BLOCKED by security control
08:54	j.smith	CONTAINMENT	Account disabled / sessions terminated

SIMULATED LAB EVIDENCE - No real patient or employer data

Figure 2. Simulated activity timeline showing sensitive-system access and the blocked download attempt.

3. Containment

After user verification, I treated the event as a likely account compromise. Recommended containment included temporarily disabling the account, terminating active sessions, initiating a credential reset, reviewing MFA, and escalating for additional log analysis.

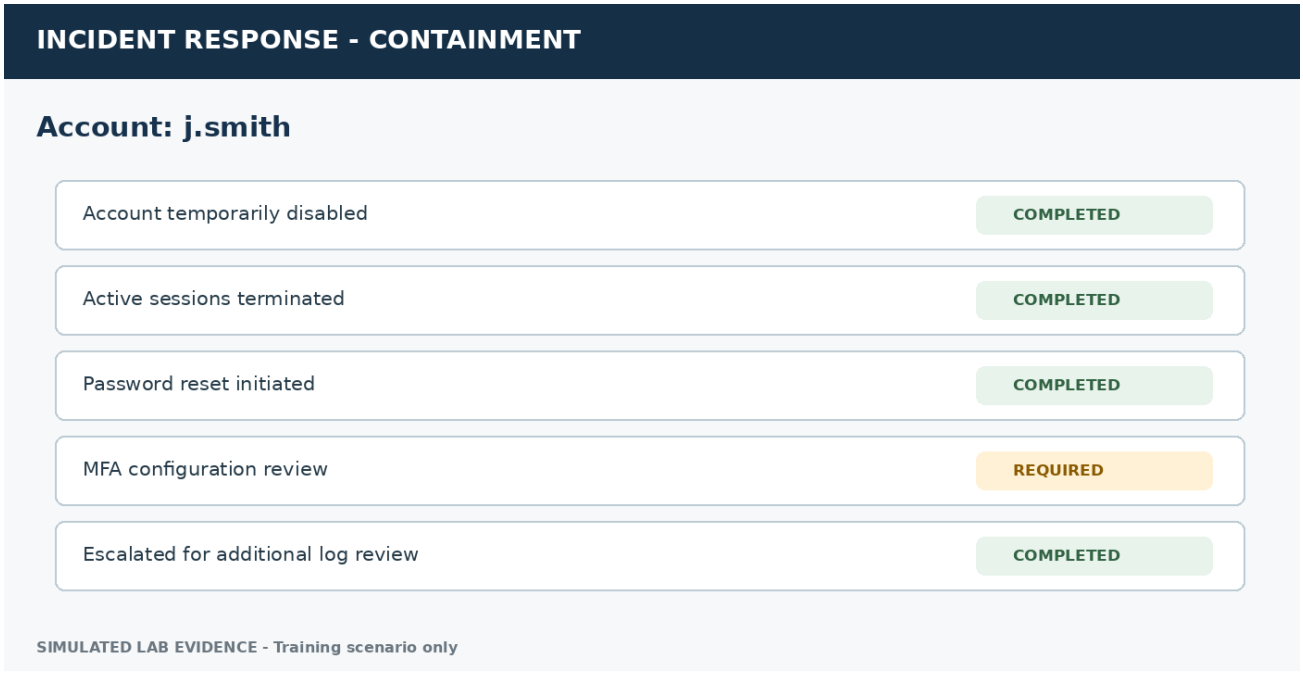


Figure 3. Simulated incident-response view documenting containment actions.

4. Impact Assessment

The unauthorized session accessed the employee portal and billing system and selected 247 billing records. A security control blocked the download. Based on the available evidence, I would report confirmed unauthorized access and an attempted download, but I would **not** claim that the 247 records were successfully exfiltrated.

5. Remediation Recommendations

• Require and verify MFA for sensitive systems and investigate any authentication path that permits access without completed MFA.

• Alert on unusual geographic logins, unknown devices, repeated failures, and impossible-travel patterns.

• Review the compromised account for additional indicators of compromise and confirm that no other sessions or credentials were affected.

• Maintain least-privilege access and controls that restrict unusual bulk downloads of sensitive records.

• Reinforce phishing, password-security, and suspicious-login reporting awareness.

6. Skills Demonstrated

SIEM alert review | Authentication-log analysis | Incident triage | Account-compromise investigation | User verification | Containment planning | Impact assessment | MFA | Least privilege | Security documentation

All screenshots in this project are simulated lab evidence created for portfolio and training purposes. No real patient, employer, or confidential information is shown.